quantity, to improve its appearance, flavor, texture or storage properties.

Food additives may be classified into two categories. Additives of the first category includes:

- Coloring agents, e.g. saffron, turmeric
- Flavoring agents, e.g. vanilla essence
- Sweeteners, e.g.saccharin
- Preservatives, e.g.ascorbic acid, sodium benzoate
- Acidity imparting agents, e.g. citric acid, acetic acid, etc.

These agents are generally considered safe for human consumption. Additives of the "Second category" are, strictly speaking, contaminants incidental through packing, processing steps, farming practices (insecticides) or other environmental conditions.

Uncontrolled or indiscriminate use of food additives may pose health hazards among consumers. For example, certain preservatives such as nitrites and nitrates can lead to the production of toxic substance, e.g. nitrosamines that have been implicated in cancer etiology.

Food Fortification

Fortification of food is a public health measure aimed at reinforcing the usual dietary intake of nutrients with additional supplies to prevent / control some nutritional disorders.

WHO has defined "Food fortification" as the process whereby nutrients are added to foods (in relatively small quantities) to maintain or improve the quality of the diet of a group, a community or a population." For example,

1. Foods artificially fortified with vitamin 'D' e.g. milk, margarine, Vanaspati and infant foods.
2. Foods fortified with vitamin 'A' e.g. margarine, milk, Vanaspati.

Adulteration of Food



Adulteration of food is an age – old problem. It is done through a large number of practices:

- Mixing
- Substitution
- Abstraction
- Concealing the quality
- Putting up decomposed foods for sale
- Misbranding or
- Giving false labels and
- Addition of toxicants.

Food adulteration practices vary from one part of the country to another, and from time to time.

The types of adulteration commonly found in India are as follows:

1. **Milk:** Addition of water, removal of fat and addition of starch to make the milk thicker are the common forms of milk adulteration.
2. **Ghee:** This is adulterated with dalda and animal fats such as pig's fat.
3. **Rice and wheat:** These are mixed with stone chips and mud to increase the bulk
4. **Flour:** Wheat flour is mixed with soap stone powder and cheaper flours such as singhada flour.
5. **Pulses:** Chemical substances are added to old stocks to improve the appearance.
6. **Tea and coffee:** Tea leaves are adulterated with old tea leaves; leather and saw dust, coffee is adulterated with chicory
7. **Honey:** This is adulterated with sugar or jaggery and boiled with empty beehives.
8. **Medicines:** Even drugs are adulterated.

The prevention of Food Adulteration Act 1954

Any food that does not confirm to the minimum standards is said to adulterated standards. Provisions have been laid down under this Act for various foods.

Unit_5_Applied_nutrition_Gen_Nur.indd 137 04-03-2020 17:12:20